

[Mr Chu's Soln] 2025 GCE O level AM P2.pdf

29 October 2025

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2025 GCE O
level AM ...



MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Ordinary Level

CANDIDATE
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ADDITIONAL MATHEMATICS

4049/02

Paper 2

October/November 2025

2 hours 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE ON ANY BARCODES.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

This document consists of 17 printed pages and 3 blank pages.



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Mathematical Formulae

1. ALGEBRA

*Quadratic Equation*For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

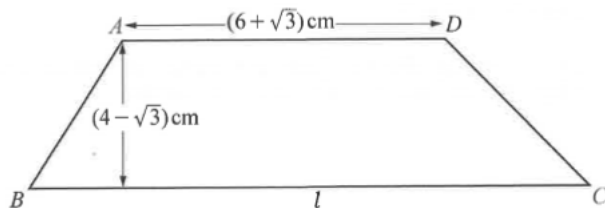
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$





1 Do not use a calculator in this question.



The diagram shows a trapezium $ABCD$ with BC and AD parallel. The length of AD is $(6 + \sqrt{3})$ cm and the perpendicular height of the trapezium is $(4 - \sqrt{3})$ cm. The length of BC is l cm. Given that the trapezium has area 26 cm^2 , find the exact length of BC . Give your answer in the form $a + b\sqrt{3}$ where a and b are integers. [6]

$$\frac{1}{2}(l + 6 + \sqrt{3})(4 - \sqrt{3}) = 26 \quad \text{--- M1}$$

$$(l + 6 + \sqrt{3})(4 - \sqrt{3}) = 52$$

$$4(l + 6) - (l + 6)\sqrt{3} + 4\sqrt{3} - 3 = 52 \quad \text{--- M1}$$

$$(4 - \sqrt{3})l = 52 + 3 - 24 + 2\sqrt{3}$$

$$l = \frac{31 + 2\sqrt{3}}{4 - \sqrt{3}} \left(\frac{4 + \sqrt{3}}{4 + \sqrt{3}} \right)$$

$$= \frac{124 + 8\sqrt{3} + 31\sqrt{3} + 6}{16 - 3} \quad \text{--- M1}$$

$$= \frac{130 + 39\sqrt{3}}{13}$$

$$= (10 + 3\sqrt{3})$$

$$BC = (10 + 3\sqrt{3}) \text{ cm.} \quad \text{--- A1}$$





- 2 On a winter's day the temperature, $T^{\circ}\text{C}$, h hours after measurement began, is modelled by the formula $T = 4 + 2h - \frac{1}{2}h^2$, where $0 \leq h \leq 8$.

(a) By expressing T in the form $A - B(h+C)^2$, where A , B and C are constants, find the time at which the temperature is greatest and state the value of this temperature. [5]

$$\begin{aligned}
 T &= -\frac{1}{2}h^2 + 2h + 4 \\
 &= -\frac{1}{2}[h^2 - 4] + 4 \quad -M1 \\
 &= -\frac{1}{2}[(h-2)^2 - 2^2] + 4 \quad -M1 \\
 &= -\frac{1}{2}(h-2)^2 + 2 + 4 \\
 &= 6 - \frac{1}{2}(h-2)^2 \quad -M1
 \end{aligned}$$

Time of greatest temperature = 2 hours after measurement begins } A1

Greatest temperature = 6°C . -A1

- (b) On the following day, and at the same times as before, the temperatures are found to be half their previous day's values. State the time at which the temperature is greatest and the value of this temperature. [2]

Time = 2 hours -A1

Greatest temperature = 3°C . -A1





- 3 The gradient of a curve, at a point (x, y) on the curve, is given by $e^x(3e^{2x} - e^{-4x}) - \cos 2x$. The curve passes through the origin. Find y in terms of x . [7]

$$\begin{aligned}\frac{dy}{dx} &= e^x(3e^{2x} - e^{-4x}) - \cos 2x \\ &= 3e^{3x} - e^{-3x} - \cos 2x \quad \text{--- M1}\end{aligned}$$

$$\begin{aligned}y &= \int 3e^{3x} - e^{-3x} - \cos 2x \, dx \\ &= \frac{3e^{3x}}{3} - \frac{e^{-3x}}{-3} - \frac{1}{2} \sin 2x + c \quad \text{--- M1} \\ &= e^{3x} + \frac{1}{3}e^{-3x} - \frac{1}{2} \sin 2x + c \quad \text{--- M1}\end{aligned}$$

When $x=0$, $y=0$.

$$0 = 1 + \frac{1}{3} + c \quad \text{--- M1}$$

$$c = -1\frac{1}{3} \quad \text{--- M1}$$

$$y = e^{3x} + \frac{1}{3}e^{-3x} - \frac{1}{2} \sin 2x - 1\frac{1}{3} \quad \text{--- A1}$$





4 Do not use a calculator in this question.

- (a) It is given that $f(x) = x^3 + px^2 + qx - 6$, where p and q are constants.
 $(x+1)$ is a factor of $f(x)$.
 When $f(x)$ is divided by $(x+2)$ there is a remainder of 4.

Find the values of p and q .

[6]

$$\begin{aligned}
 f(-1) &= 0 \\
 (-1)^3 + p(-1)^2 + q(-1) - 6 &= 0 \quad \text{--- M1} \\
 -1 + p - q - 6 &= 0 \\
 p - q &= 7 \quad \text{--- (1) --- M1} \\
 f(-2) &= 4 \\
 (-2)^3 + p(-2)^2 + q(-2) - 6 &= 4 \quad \text{--- M1} \\
 -8 + 4p - 2q - 6 &= 4 \\
 2p - q &= 9 \quad \text{--- (2) --- M1} \\
 (2) - (1): \quad p &= 2 \quad \text{--- A1} \\
 q &= -5 \quad \text{--- A1}
 \end{aligned}$$

- (b) It is given that $3x-1$ is a factor of $g(x) = 9x^3 - 18x^2 - x + 2$. Express $g(x)$ as a product of three linear factors. [3]

$$\begin{aligned}
 9x^3 - 18x^2 - x + 2 &= (3x-1)(3x^2 + Kx - 2), \text{ K is constant to be found} \\
 \text{compare } x^2: \quad -18 &= -3 + 3K \\
 -15 &= 3K \\
 K &= -5 \\
 g(x) &= (3x-1)(3x^2 - 5x - 2) \quad \text{--- M1} \\
 &= (3x-1)(3x+1)(x-2) \\
 &\quad \text{--- A1} \quad \text{--- A1}
 \end{aligned}$$





5 Given that $y = \frac{1-e^{-x}}{1+e^{-x}}$, prove that y increases for all real values of x .

[6]

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{\overbrace{(1+e^{-x})(e^{-x})}^{M1} - \underbrace{(1-e^{-x})(-e^{-x})}_{-M1}}{(1+e^{-x})^2 - M1} \\
 &= \frac{e^{-x} + e^{-2x} + e^{-x} - e^{-2x}}{(1+e^{-x})^2} \\
 &= \frac{2e^{-x}}{(1+e^{-x})^2} \quad \text{M1}
 \end{aligned}$$

since $e^{-x} > 0$ for all real values of x , M1
 $2e^{-x} > 0$ and $(1+e^{-x})^2 > 0$

$\therefore \frac{dy}{dx} > 0$ for all real values of x A1
 and y is an increasing function





6 Solve the equation $2 \operatorname{cosec}^2 \theta - \cot \theta - 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

[7]

$$2(1 + \cot^2 \theta) - \cot \theta - 3 = 0 \quad \text{--- M1}$$

$$2 \cot^2 \theta - \cot \theta - 1 = 0$$

$$(2 \cot \theta + 1)(\cot \theta - 1) = 0 \quad \text{--- M1}$$

$$\cot \theta = -\frac{1}{2} \quad \text{or} \quad \cot \theta = 1$$

$$\tan \theta = -2 \quad \text{or} \quad \tan \theta = 1 \quad \text{--- M1}$$

$$\alpha_1 = \tan^{-1}(2) \\ = 63.4^\circ \quad \text{--- M1}$$

$$\theta = 116.6^\circ \quad \text{or} \quad 296.6^\circ$$

✓
A1

$$\alpha_2 = \tan^{-1}(1) = 45^\circ \quad \text{--- M1}$$

$$\theta = 45^\circ \quad \text{or} \quad 225^\circ$$

✓
A1

$$\theta = 45^\circ \quad \text{or} \quad 116.6^\circ \quad \text{or} \quad 225^\circ \quad \text{or} \quad 296.6^\circ$$





- 7 A particle is travelling in a straight line. Its acceleration, $a \text{ m/s}^2$, at time t seconds is given by

$$a = (2t+1)^{-1} + 0.3t + 0.2 \text{ for } 0 \leq t \leq 10.$$

At $t = 0$ the velocity of the particle is -3 m/s .

Find the velocity of the particle when $t = 5$.

[7]

$$a = \frac{1}{2t+1} + 0.3t + 0.2$$

$$v = \int \frac{1}{2t+1} + 0.3t + 0.2 \, dt \quad \text{--- M1}$$

$$= \frac{1}{2} \ln(2t+1) + 0.15t^2 + 0.2t + c \quad \text{--- M1}$$

$$\text{when } t=0, \, v = -3$$

$$-3 = c \quad \text{--- M1}$$

$$v|_{t=5} = \frac{1}{2} \ln(10+1) + 0.15(5)^2 + 0.2(5) - 3 \quad \text{--- M1}$$

$$= 2.95 \text{ m/s} \quad \text{--- A1}$$



8 (a) Solve the equation $2^p = 5(3^{2-p})$.

[4]

$$2^p = 5(3^2 \times 3^{-p})$$

$$2^p = 45(3^{-p}) \text{ --- M1}$$

$$2^p \times 3^p = 45$$

$$6^p = 45 \text{ --- M1}$$

$$p(\ln 6) = \ln 45 \text{ --- M1}$$

$$p = \frac{\ln 45}{\ln 6}$$

$$= 2.12 \text{ (3 s.f.) --- A1}$$





(b) Solve the simultaneous equations

$$\begin{aligned} 9(2^x) + 2(3^{y+1}) &= 38, \\ 15(2^{x-1}) - 21(3^y) &= 23. \end{aligned}$$

[5]

$$\begin{aligned} 9(2^x) + 6(3^y) &= 38 \quad (1) \\ 7.5(2^x) - 21(3^y) &= 23 \quad (2) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{M1}$$

$$\begin{aligned} (1) \times 7: \quad 63(2^x) + 42(3^y) &= 266 \quad (3) \\ (2) \times 2: \quad 15(2^x) - 42(3^y) &= 46 \quad (4) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{M1: attempt to eliminate or substitute}$$

$$(3) + (4): \quad 78(2^x) = 312 \quad \text{--- M1}$$

$$2^x = 4$$

$$x = 2 \quad \text{--- A1}$$

$$15(2^2) - 42(3^y) = 46$$

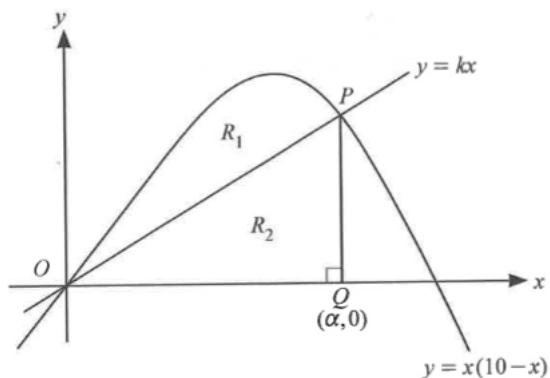
$$-42(3^y) = -14$$

$$3^y = \frac{1}{3}$$

$$y = -1 \quad \text{--- A1}$$

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The diagram shows the line $y = kx$, where k is a constant, $k < 10$, and the curve $y = x(10-x)$. The line intersects the curve at the origin O and at point P . Point $Q(\alpha, 0)$, where α is a constant, lies on the x -axis such that PQ is parallel to the y -axis.

R_1 is the area of the region enclosed by the curve and the line OP .

- (a) By finding k in terms of α , show that $R_1 = \frac{\alpha^3}{6}$.

[7]

$$kx = x(10-x) \quad -M1$$

solution is $x = \alpha$. Hence,

$$k\alpha = \alpha(10-\alpha).$$

$$k = 10 - \alpha \quad (\alpha \neq 0). \quad -M1$$

$$R_1 = \int_0^\alpha [x(10-x) - kx] dx \quad -M1$$

$$= \int_0^\alpha (10x - x^2 - kx) dx$$

$$= \left[\frac{(10-k)x^2}{2} - \frac{x^3}{3} \right]_0^\alpha$$

$$= \frac{(10-k)\alpha^2}{2} - \frac{\alpha^3}{3} \quad -M1$$

$$= \frac{(10-(10-\alpha))\alpha^2}{2} - \frac{\alpha^3}{3} = \frac{\alpha^3}{2} - \frac{\alpha^3}{3} = \frac{\alpha^3}{6} \quad -A1$$





R_2 is the area of the region enclosed by the lines OP , PQ and the x -axis.

(b) Find the numerical value of k if $R_1 = R_2$.

[3]

$$PQ = k\alpha$$

$$= \alpha(10-\alpha)$$

$$R_2 = \frac{1}{2}(\alpha)[\alpha(10-\alpha)]$$

$$\frac{1}{2}\alpha^2(10-\alpha) = \frac{\alpha^3}{6} \quad -M1$$

$$5\alpha^2 - \frac{\alpha^3}{2} = \frac{\alpha^3}{6}$$

$$5\alpha^2 - \frac{2\alpha^3}{3} = 0.$$

$$\alpha^2(5 - \frac{2}{3}\alpha) = 0 \quad -M1$$

$$\alpha = 0 \quad \text{or} \quad \alpha = 7.5$$

(neg.)

$$k = 10 - 7.5$$

$$= 2.5 \quad -A1$$

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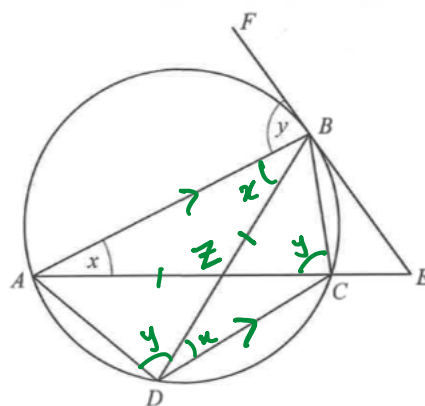
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The diagram shows a quadrilateral $ABCD$ whose vertices lie on the circumference of a circle. The tangent to the circle at B meets the line AC extended at the point E . Angle $BAC = x$ and angle $FBA = y$, where F is a point on EB extended.

(a) Prove that triangle ABE is similar to triangle BCE .

[3]

$$\angle AEB = \angle BEC \text{ (common angle) } -M1$$

$$\angle BAE = \angle CBE \text{ (tangent chord theorem) } -M1$$

$$\therefore \triangle ABE \text{ similar to } \triangle BCE \text{ (AA similarity test) } -A1$$





(b) (i) Prove that angle $ADC = x + y$.

[3]

$$\angle ADB = y \text{ (tangent chord theorem) - M1}$$

$$\angle BDC = x \text{ (angles in same segment) - M1}$$

$$\begin{aligned} \therefore \angle ADC &= \angle ADB + \angle BDC \\ &= x + y \quad \text{- A1} \end{aligned}$$

(ii) If DC is parallel to AB prove that $AD = BC$.

[4]

Let AC and BD intersect at Z

$$\angle ADZ = \angle BCZ \text{ (angles in same segment) [A] - M1}$$

$$\angle AZD = \angle BZC \text{ (vertically opposite angles) [A] - M1}$$

$$\begin{aligned} \angle ABZ &= \angle BDC \text{ (alternate angles, } DC \parallel AB) \\ &= \angle BAC \text{ (from (i))} \end{aligned} \quad \left. \vphantom{\begin{aligned} \angle ABZ &= \angle BDC \\ &= \angle BAC \end{aligned}} \right\} \text{M1}$$

Hence, $\triangle ABZ$ is isosceles triangle

$$\text{with } AZ = BZ \quad \text{[S]}$$

Hence $\triangle ADZ$ and $\triangle BCZ$ are congruent (AAS congruency test)

$$\therefore AD = BC$$





- 11 At time $t = 0$ seconds, two objects are at points A and B which are 30 cm apart. The objects are moving directly towards one another in a straight line. Their constant speeds are $6 \cos \theta$ cm/s and $8 \sin \theta$ cm/s respectively, for a certain value of θ measured in degrees. The radii of the objects may be ignored.

(a) Explain why θ must be acute. [1]

θ must be acute or else $6 \cos \theta$ will be negative and the objects will not be moving towards one another.

- (b) In the case when the objects collide at the midpoint of AB , find the time taken for the objects to collide. [3]

$$\begin{aligned} \text{Time taken by A} &= \frac{15}{6 \cos \theta} \text{ s} \\ &= \frac{5}{2 \cos \theta} \text{ s} \end{aligned}$$

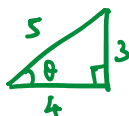
$$\text{Time taken by B} = \frac{15}{8 \sin \theta} \text{ s}$$

$$\frac{5}{2 \cos \theta} = \frac{15}{8 \sin \theta} \text{ — M1}$$

$$40 \sin \theta = 30 \cos \theta$$

$$\tan \theta = \frac{3}{4} \text{ — M1}$$

$$\cos \theta = \frac{4}{5}$$



$$\begin{aligned} \text{Time taken} &= \frac{5}{2 \left(\frac{4}{5} \right)} \\ &= \frac{25}{8} = 3.125 \text{ s. — A1} \end{aligned}$$

- (c) (i) If, instead, the objects collide when $t = 4$ seconds show that $3 \cos \theta + 4 \sin \theta = 3.75$. [2]

$$6 \cos \theta (4) + 8 \sin \theta (4) = 30 \text{ — M1}$$

$$24 \cos \theta + 32 \sin \theta = 30$$

$$\div 8: \quad 3 \cos \theta + 4 \sin \theta = 3.75 \text{ — A1}$$





- (ii) By writing $3 \cos \theta + 4 \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and α is acute, find the numerical values of the speeds of the two objects. [6]

$$R = \sqrt{3^2 + 4^2}$$

$$= 5 \quad \text{--- MI}$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$= 53.1^\circ \quad \text{--- MI}$$

$$3 \cos \theta + 4 \sin \theta = 5 \cos(\theta - 53.130^\circ) \quad \text{--- MI}$$

$$5 \cos(\theta - 53.130^\circ) = 3.75$$

$$\theta = \cos^{-1}\left(\frac{3.75}{5}\right) + 53.130^\circ \quad \text{or} \quad \underbrace{-\cos^{-1}\left(\frac{3.75}{5}\right)}_{\text{4th quadrant angle}} + 53.130^\circ$$

$$= 94.539^\circ \quad \text{or} \quad 11.720^\circ \quad \text{--- MI}$$

(reject)

$$\text{Object A speed} = 6 \cos 11.720^\circ$$

$$= 5.87 \text{ cm/s (3 sf)} \quad \left. \vphantom{\begin{matrix} 6 \cos 11.720^\circ \\ 5.87 \text{ cm/s (3 sf)} \end{matrix}} \right\} \text{AI}$$

$$\text{Object B speed} = 8 \sin 11.720^\circ$$

$$= 1.63 \text{ cm/s (3 sf)} \quad \left. \vphantom{\begin{matrix} 8 \sin 11.720^\circ \\ 1.63 \text{ cm/s (3 sf)} \end{matrix}} \right\} \text{AI}$$

